ChemComm

COMMUNICATION

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Chem. Commun.*, 2024, **60**, 12537

Received 27th August 2024,
Accepted 30th September 2024

DOI: 10.1039/d4cc04307c

rsc.li/chemcomm

Published on 09 October 2024. Downloaded by Fudan University on 2/5/2025 4:17:29 AM.

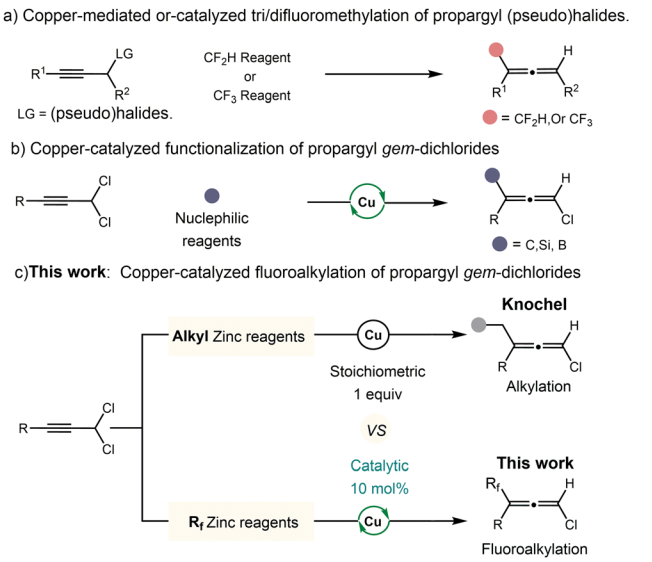
Copper-catalyzed perfluoroalkylation of propargyl *gem*-dichlorides†

Bosheng Liu,[a] Jinxu Dong,[a] Qi Zhu,[a] Jundong Zhang,[a] Jian Huang,[a] Shiwen Liu[b] and Xiaojun Zeng*[a]

Herein, we present a highly efficient copper-catalyzed protocol for transforming propargyl *gem*-dichlorides into the corresponding chloro-substituted fluoroalkylated allenes. This protocol demonstrates a broad substrate scope and excellent tolerance towards various functional groups. Moreover, the strategy of utilizing the chloro-substituted pentafluoroethylation allenes for multiple transformations has shown significant value in synthetic chemistry.

Given the unique properties and reactivity of the cumulated diene system, allenes hold significant importance in organic synthesis, serving as versatile intermediates in functional group transformations.[1] Moreover, allenes are prevalent in natural products, bioactive small molecules, and functional materials.[2] It is widely recognized that incorporating a fluoroalkyl group into pharmaceutical candidates often enhances various medicinal properties, such as significantly influencing binding affinity, improving bioavailability, enhancing membrane permeability, and providing protection against *in vivo* metabolism.[3] Therefore, the development of efficient methods for synthesizing diverse fluoroalkylated allenes[4] is evidently of great value. These methods facilitate scaffold modification to meet the evolving demands of drug discovery and material development.

Modern approaches for the synthesis of fluoroalkylated allenes typically involve copper-mediated or -catalyzed cross-coupling of propargylic electrophiles with appropriate fluoroalkyl reagents. For instance, Qing,[5] Szabo,[6] Xiao,[7] and Nishibayashi[8] independently report copper-mediated or -catalyzed trifluoromethylation of various propargylic electrophiles, including propargylic halides, acetates, and methanesulfonates, using appropriate nucleophilic CF_3 reagents to afford trisubstituted trifluoromethylallenes. Along this line, Shen[9] developed a copper catalyzed difluoromethylation of propargyl bromides using their invented CF_2H reagents. However, existing methods often require stoichiometric amounts of copper. Additionally, fluoroalkylated allenes bearing only alkyl or aryl groups lack versatility for further transformations, thereby limiting the diversity of the synthesized fluoroalkylated allene scaffolds. Moreover, to the best of our knowledge, approaches to perfluoroalkylated (*e.g.*, C_nF_{n+1}, $n \geq 2$) allenes are underexplored, despite the perfluoroalkyl moiety also possessing some unique properties.[10] Given the widespread application of cross electrophile-coupling strategies, chloro-substituted allenes serve as versatile coupling partners,[11] providing facile access to diverse allene scaffolds. Therefore, efforts to synthesize more active chlorinated fluoroalkylated allene products are desirable. Propargyl *gem*-dichlorides (Scheme 1b) have garnered significant attention from chemists due to their



Scheme 1 Research background and current work. (a) Copper-mediated or-catalyzed tri/difluoromethylation of propargyl (pseudo)halides. (b) Copper-catalyzed functionalization of propargyl *gem*-dichlorides. (c) This work.

[a] *School of Chemistry and Chemical Engineering, Nanchang University, Nanchang, Jiangxi, 330031, China. E-mail: xjzeng@ncu.edu.cn*
[b] *College of Textiles and Clothing, Yancheng Institute of Technology, Jiangsu, 224003, China*

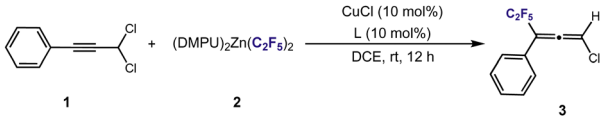
† Electronic supplementary information (ESI) available. See DOI: https://doi.org/10.1039/d4cc04307c

Published on 09 October 2024. Downloaded by Fudan University on 2/5/2025 4:17:29 AM.

versatile activity. The resulting product, containing a chlorine atom, holds particular interest as it allows for further transformations compared to allenes bearing only alkyl or aryl groups. For example, the Alexakis group[12] reported copper-catalyzed 1,3-substitution of propargyl *gem*-dichloride substrates with readily available nucleophilic Grignard reagents, yielding chlorinated allenes. Following this research direction, the Toh/Xu group[13] employed propargyl *gem*-dichlorides to synthesize chloro-substituted allenylsilanes in the presence of a copper catalyst using a compatible $PhMe_2Si$-Bpin reagent. More recently, the Xu group[14] demonstrated that catalytic copper enables the borylation of propargyl *gem*-dichlorides for the synthesis of chlorosubstituted allenyl-Bdan products. Despite the growing interest in preparing various functional chlorinated allenes, methods to achieve chloro-substituted fluoroalkylated allenes remain relatively underexplored. Inspired by the pioneering work of the Knochel group[15] (Scheme 1c) on propargyl *gem*-dichlorides for copper-mediated cross-coupling reactions with alkyl zinc reagents, we were motivated to investigate the possibility of a copper-catalyzed perfluoroalkylation of propargyl *gem*-dichlorides using suitable $(DMPU)_2Zn(R_f)_2$ reagents.

To test the validity of this hypothesis, propargyl *gem*-dichloride (**1**), prepared from the corresponding propargylic aldehyde, was selected as the model substrate for investigating the reaction conditions (Table 1). Following a thorough optimization campaign, it was determined that in the presence of catalytic amounts of CuCl (10 mol%), phenanthroline ligand L_3 (10 mol%), and 0.8 equiv. of znic-C_2F_5 reagent **2**, facilitated the effective transformation of compound **1** into the chloro-substituted pentafluoroethylated allenes **3**. This transformation was achieved with an impressive yield of 92% at room temperature in 1,2-dichloroethane (DCE) (entry 1). The use of the other copper salts such as Cu(I) or Cu(II), resulted in a significantly decreased yield (entries 2 and 3), while replacing the copper catalyst with a nickel catalyst did not yield desirable products (entry 4). Further exploration into the commercially available nucleophilic C_2F_5 reagents revealed that the *in situ* generation of CuC_2F_5 species in the presence of $TMSC_2F_5$ was slower compared to the use of $(DMPU)_2Zn(C_2F_5)_2$, which exhibited superior reactivity (entry 5). The reaction had deleterious effects on the coupling reactions when equal amounts of the C_2F_5 source were used (entry 6). Among the commercially available ligands, 2,9-dimethyl-1,10-phenanthroline ligand proved to be the most effective, while other ligands, including bipyridine or tridentate ligands (2,2′:6′,2″-terpyridine) were less effective (entries 7 and 8). Solvents play an important role in this pentafluoroethylation reaction; replacing DCE with other solvents led to diminished yields (entries 9–11). Control reactions supported our proposed mechanism, with no conversion occurring in the absence of Cu or ligand (entries 12 and 13). Interestingly, even when the reaction was conducted in an open-air environment, only a minor decrease in the yield was observed (entry 13).

Table 1 Screening the reaction conditions[a]



1 2 3

Entry	Variations from standard conditions	Yield (3)[b] (%)
1	dmphen as ligand	92
2	CuOTf as the catalyst	52
3	CuF_2 as the catalyst	10
4	$NiCl_2$ glyme as the catalyst	Trace[c]
5	$TMSCF_2CF_3$ instead of **2**	Trace[d]
6	0.5 equiv. of **2**	5
7	Terpyridine as ligand	85
8	Bipyridine as ligand	83
9	DMSO instead of DCE	45
10	Toluene instead of DCE	21
11	THF instead of DCE	Trace
12	No CuI	0
13	No ligand	Trace
14	Reaction set up in air	86

[a] Unless otherwise noted, the reaction was carried out with **1** (0.2 mmol), **2** (0.16 mmol), CuCl catalyst (0.01 mmol), and dmphen (2,9-dimethyl-1,10-phenanthroline 0.01 mmol) in solvent DCE (1.0 mL) at rt for 12 h. [b] Yields determined by ^{19}F NMR using $PhCF_3$ as an internal standard. [c] The 0% mol $NiCl_2$ glyme and 10% mol L1 were used. [d] The 0.16 mmol $TMSCF_2CF_3$ were used.

With these optimized conditions established, a broad range of propargyl *gem*-dichlorides were used as an electrophile to install a pentafluoroethyl group on the chloroallenic moiety (Table 2). Substrates featuring diverse aryl groups were efficiently converted into the respective chloro-substituted pentafluoroethylated allenes, yielding high amounts of product quantities with notable selectivity (**3–16**). Generally, as the electron density on the aromatic ring increased, the efficiency of this transformation tended to rise. For example, the *ortho*-substituted electron-rich groups, such as methyl **4** and alkoxy (**5**, **6**), exhibited higher yields compared to electron-deficit groups like cyano **7** and nitro **8**. However, aryl substitution with an electron-donating group at the *para* position furnished the corresponding products in lower yields (**11**, **12**). A diverse range of synthetic handles showed good tolerance to the reaction, including ketone **13**, and ester **14**. Substrates bearing halogens such as Cl and Br (**9**, **10**, **15**) were well accommodated, enabling further modification of pentafluoroethylation products. Moreover, the current copper-catalyzed process shows the much larger substrate scope than the previous copper catalyzed process,[14b] where the aryl substituted propargyl *gem*-dichlorides led to no reported results.

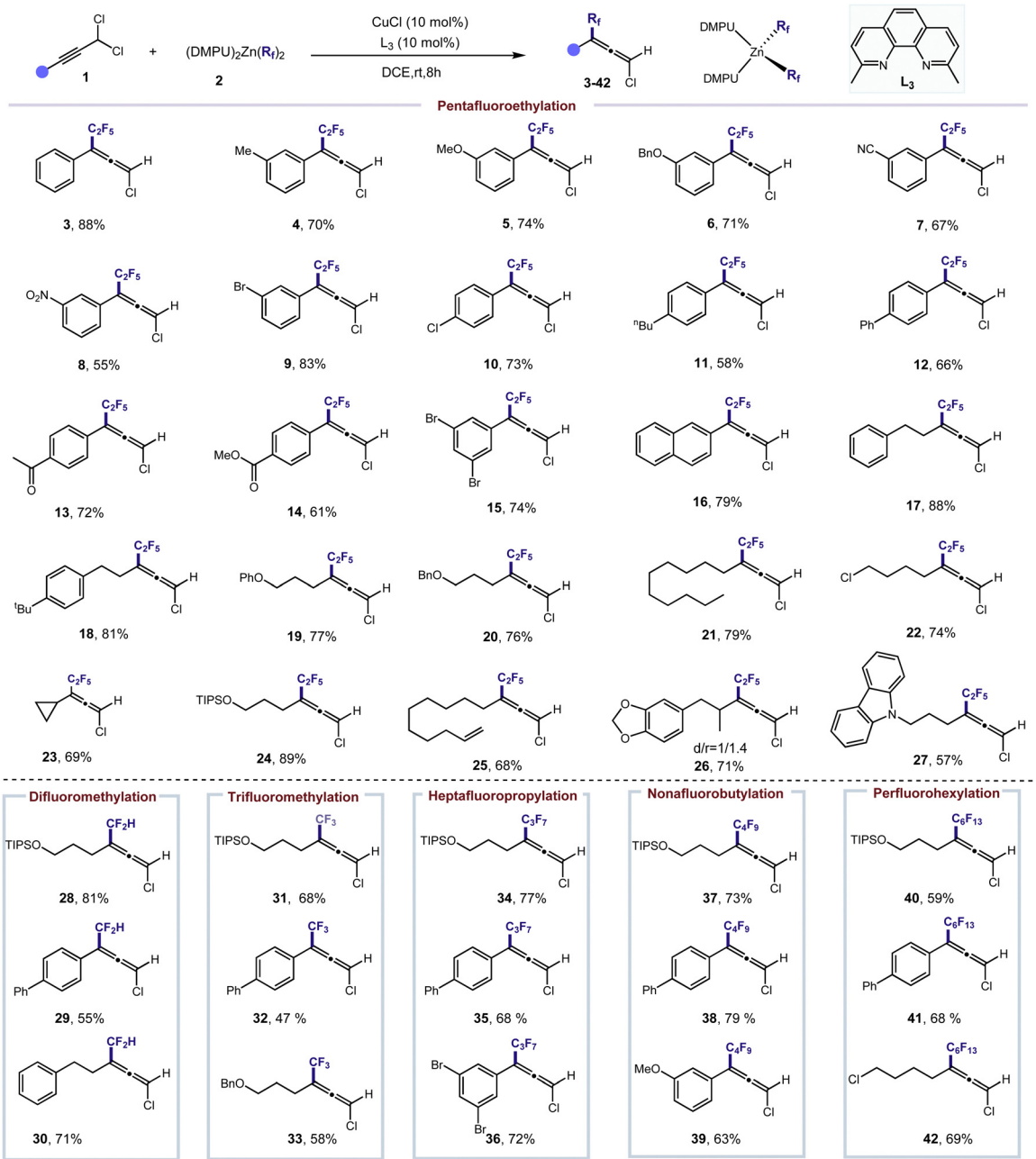
Next, we explored the substrate scope of this reaction by using a series of alkyl substituted propargyl *gem*-dichlorides. It was observed that substrates containing electron-rich alkylbenzene rings had a minimal effect on the reaction (**17**, **18**). As expected, substrates with different primary alkyl groups, such as *n*-decanyl **21**, chloride **22** and alkene **25**, exerted a deleterious effect on the reaction. However, substrates with functional groups such as phenol **19**, silyl ether **24**, acetal **26**, and carbazole **27** groups were all compatible to afford the corresponding products. Remarkably, pentafluoroethylation of a radical clock cyclopropane substituent *gem*-dichloride afforded the desirable product **23** rather than the ring opening product, indicating that this process did not involve a radical intermediate.

Given the value of CF_2H as a bioisostere of the thiol and amino groups,[16] we proceeded to investigate the extension of this reaction to the difluoromethylation of the propargyl

 This journal is © The Royal Society of Chemistry 2024

Published on 09 October 2024. Downloaded by Fudan University on 2/5/2025 4:17:29 AM.

Table 2 Substrate scope of the copper catalyzed perfluoroalkylation of propargyl *gem*-dichlorides[a]

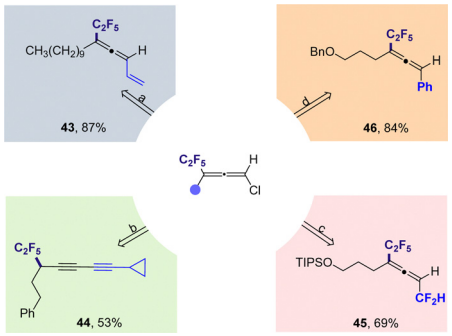


[a] Reactions were run with 0.20 mmol of propargyl *gem*-dichloride (1.0 equiv.), 0.16 mmol of $(DMPU)_2Zn(R_f)_2$ (0.8 equiv.), CuCl (10 mol%) and phen (10 mol%) in DCE (1 mL) at rt for 8 h. The isolated yield is provided.

gem-dichlorides using the Vicic–Mikami[17] CF_2H reagent. Substrates bearing aromatic and aliphatic groups smoothy delivered adducts **28–30** with good efficiency (55–81% yield). Furthermore, we investigated whether this method could be expanded to install longer-chain perfluoroalkyl groups. To achieve this, $(DMPU)_2Zn(C_nF_{2n+1})_2$[18] were employed as compatible R_f reagents, yielding the corresponding chloro-substituted fluoroalkylated allenes with useful efficiencies. Trifluoromethylation (**31–33**), heptafluoropropylation (**34–36**), nonafluorobutylation (**37–39**) and perfluorohexylation (**40–42**) effectively demonstrated the versatility of this protocol, representing a novel and straightforward approach for introducing diverse perfluoroalkyl groups onto chloro-substituted allenes.

To further showcase the potential utility of the synthesized chloro-substituted fluoroalkylated allenes, their subsequent conversion into diverse fluoroalkylated compounds can be achieved through simple one-step operation (Scheme 2). For example, the chloro-substituted allene **21** could be utilized to produce pentafluoroethylated compound **43** *via* Suzuki coupling reaction with potassium vinyltrifluoroborate. Similarly, the

Published on 09 October 2024. Downloaded by Fudan University on 2/5/2025 4:17:29 AM.



Scheme 2 Synthetic applications. Reaction conditions: please see ESI.†

chlorinated allenes **17** could be subjected to a Sonogashira coupling reaction, resulting in the production of 1,3-diyne products **44**. Additionally, chlorinated allenes **24** can be transformed into the corresponding difluoromethylated **45** through a copper-catalyzed coupling reaction. Furthermore, the palladium-catalyzed coupling reaction of chlorinated allenes **27** with phenylboronic acid provided compound **46** in good yield.

Based on the presented results and literature reports,[19] we proposed a possible mechanism (please see ESI,† Scheme S1). The [Cu^1-R_f] intermediate is hypothesized to be generated *in situ* through a transmetalation between $(DMPU)_2Zn(R_f)_2$ reagent and Cu^I(I). Subsequently, in the presence of propargyl *gem*-dichlorides, a π-complex intermediate **B** was formed *in situ*. Following this, a divergent pathway may occur: either a fluoroalkyl copper anion attack on the acetylene carbon, leading to an S_N2'[20] substitution to form copper(III) intermediate **C**, or an anti-selective addition occurred, resulting in copper(I) intermediate **D**. Subsequently, reductive elimination from **C** or β-elimination from **D** yields the target chloro-substituted fluoroalkylated allenes **3**, with the copper catalyst being released for the next catalytic cycle.

In conclusion, we have developed a novel method for the copper-catalyzed cross-coupling of propargyl *gem*-dichlorides with $(DMPU)_2Zn(R_f)_2$ to assemble chloro-substituted fluoroalkylated allenes. The reaction proceeded under mild conditions and exhibited good tolerance towards various functional groups. The introduction of fluoroalkyl onto chlorinated allenes, allowing transition metal-catalyzed cross-coupling, serves as an important scaffold to be further transformed to trisubstituted allenes. Due to its simplicity, this methodology demonstrates broad applicability in the preparation of molecules containing fluoroalkylated groups.

We gratefully acknowledge the financial support from the National Natural Science Foundation of China (No. 22101111, 22361027), Natural Science Foundation of Chongqing (2024NSCQ-MSX4604), Jiangxi Provincial Natural Science Foundation (20242 BAB23017), and Double Thousand Plan of Jiangxi Province (JXSQ2023101071).

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

Notes and references

1 (*a*) S. Yu and S. Ma, *Angew. Chem., Int. Ed.*, 2012, **51**, 3074–3112; (*b*) S. Ma, *Chem. Rev.*, 2005, **105**, 2829–2872; (*c*) L. Liu, R. M. Ward and J. M. Schomaker, *Chem. Rev.*, 2019, **119**, 12422–12490; (*d*) D.-D. Hu, Q. Gao, J.-C. Dai, R. Cui, Y.-B. Li, Y.-M. Li, X.-G. Zhou, K.-J. Bian, B.-B. Wu, K.-F. Zhang, X.-S. Wang and Y. Li, *Sci. China: Chem.*, 2022, **65**, 753–761; (*e*) K.-F. Zhang, K.-J. Bian, C. Li, J. Sheng, Y. Li and X.-S. Wang, *Angew. Chem., Int. Ed.*, 2019, **58**, 5069–5074.
2 (*a*) P. Rivera-Fuentes and F. Diederich, *Angew. Chem., Int. Ed.*, 2012, **51**, 2818–2828; (*b*) A. Hoffmann-Röder and N. Krause, *Angew. Chem., Int. Ed.*, 2004, **43**, 1196–1216.
3 (*a*) W. K. Hagmann, *J. Med. Chem.*, 2008, **51**, 4359–4369; (*b*) S. Purser, P. R. Moore, S. Swallow and V. Gouverneur, *Chem. Soc. Rev.*, 2008, **37**, 320–330; (*c*) Z. Jin, F. Zhang, X. Xiao, N. Wang, X. Lv and L. Zhou, *Org. Chem. Front.*, 2024, **11**, 2112–2133.
4 (*a*) M. Taj Muhammad, Y. Jiao, C. Ye, M.-F. Chiou, M. Israr, X. Zhu, Y. Li, Z. Wen, A. Studer and H. Bao, *Nat. Commun.*, 2020, **11**, 416; (*b*) S. Akiyama, S. Nomura, K. Kubota and H. Ito, *J. Org. Chem.*, 2020, **85**, 4172–4181; (*c*) B. R. Ambler, S. Peddi and R. A. Altman, *Org. Lett.*, 2015, **17**, 2506–2509; (*d*) S. Qi, S. Gao, X. Xie, J. Yang and J. Zhang, *Org. Lett.*, 2020, **22**, 5229–5234; (*e*) C. Yang, Z.-L. Liu, D.-T. Dai, Q. Li, W.-W. Ma, M. Zhao and Y.-H. Xu, *Org. Lett.*, 2020, **22**, 1360–1367; (*f*) M. Deliaval, R. Jayarajan, L. Eriksson and K. J. Szabó, *J. Am. Chem. Soc.*, 2023, **145**, 10001–10006; (*g*) J.-L. Zhang, J.-X. Xiong, L.-Q. Fei, F.-H. Cui, S.-H. Li, H.-T. Tang, Y.-Z. Xie and Y.-M. Pan, *Org. Chem. Front.*, 2024, DOI: 10.1039/d4qo00332b; (*h*) X. Li, N. Li, L. Yang, R. Jiang, Y. He, J. Shi, Z.-X. Jiang and Z. Yang, *ACS Catal.*, 2023, **13**, 12755–12765; (*i*) W. Liu, C. Liu, M. Wang and W. Kong, *ACS Catal.*, 2022, **12**, 10207–10221; (*j*) H.-J. Miao, J.-H. Zhang, S. Liu, W.-H. Wang, X. Yang, X.-H. Duan and L.-N. Guo, *Org. Lett.*, 2023, **25**, 5563–5568.
5 X. Jiang and F.-L. Qing, *Beilstein J. Org. Chem.*, 2013, **9**, 2862–2865.
6 T. S. N. Zhao and K. J. Szabó, *Org. Lett.*, 2012, **14**, 3966–3969.
7 Y.-L. Ji, J.-J. Kong, J.-H. Lin, J.-C. Xiao and Y.-C. Gu, *Org. Biomol. Chem.*, 2014, **12**, 2903–2906.
8 Y. Miyake, S.-i Ota, M. Shibata, K. Nakajima and Y. Nishibayashi, *Chem. Commun.*, 2013, **49**, 7809–7811.
9 Y. Gu, C. Lu, Y. Gu and Q. Shen, *Chin. J. Chem.*, 2018, **36**, 55–58.
10 (*a*) J. P. Sorrentino and R. A. Altman, *ACS Med. Chem. Lett.*, 2022, **13**, 707–713; (*b*) E. Prchalová, O. Štěpánek, S. Smrček and M. Kotora, *Future Med. Chem.*, 2014, **6**, 1201–1229.
11 Q. Ran, K.-F. Wu and Y.-H. Xu, *Org. Lett.*, 2024, **26**, 3767–3771.
12 (*a*) H. Li, D. Grassi, L. Guénée, T. Bürgi and A. Alexakis, *Chem. – Eur. J.*, 2014, **20**, 16694–16706; (*b*) H. Li, D. Müller, L. Guénée and A. Alexakis, *Org. Lett.*, 2012, **14**, 5880–5883.
13 Z.-L. Liu, C. Yang, Q.-Y. Xue, M. Zhao, C.-C. Shan, Y.-H. Xu and T.-P. Loh, *Angew. Chem., Int. Ed.*, 2019, **58**, 16538–16542.
14 (*a*) K.-F. Wu, D.-T. Dai, X.-Y. Sun and Y.-H. Xu, *Org. Lett.*, 2022, **24**, 2660–2664; (*b*) Q. Ran, K.-F. Wu and Y.-H. Xu, *Org. Lett.*, 2024, DOI: 10.1021/acs.orglett.4c00864.
15 M. A. Schade, S. Yamada and P. Knochel, *Chem. – Eur. J.*, 2011, **17**, 4232–4237.
16 Y. Zafrani, D. Yeffet, G. Sod-Moriah, A. Berliner, D. Amir, D. Marciano, E. Gershonov and S. Saphier, *J. Med. Chem.*, 2017, **60**, 797–804.
17 (*a*) L. Xu and D. A. Vicic, *J. Am. Chem. Soc.*, 2016, **138**, 2536–2539; (*b*) H. Serizawa, K. Ishii, K. Aikawa and K. Mikami, *Org. Lett.*, 2016, **18**, 3686–3689.
18 (*a*) S. Fan, C. Zheng, K. Zheng, J. Li, Y. Liu, F. Yan, H. Xiao, Y.-S. Feng and Y.-Y. Zhu, *Org. Lett.*, 2021, **23**, 3190–3194; (*b*) K. Aikawa, Y. Nakamura, Y. Yokota, W. Toya and K. Mikami, *Chem. – Eur. J.*, 2015, **21**, 96–100; (*c*) X. Zeng, W. Yan, M. Paeth, S. B. Zacate, P.-H. Hong, Y. Wang, D. Yang, K. Yang, T. Yan, C. Song, Z. Cao, M.-J. Cheng and W. Liu, *J. Am. Chem. Soc.*, 2019, **141**, 19941–19949.
19 J.-M. Dollat, J.-L. Luche and P. Crabbé, *J. Chem. Soc., Chem. Commun.*, 1977, 761–762.
20 (*a*) H. Wu, Z. Zheng, K. Zhang, J. Kajanus, M. J. Johansson, A. Córdova and J.-E. Bäckvall, *Angew. Chem., Int. Ed.*, 2023, **62**, e202314512; (*b*) R. Li, F. Ouyang, Y. Bai, R. Tang, G. Yu and B. Wei, *Org. Lett.*, 2023, **25**, 2543–2547.

 This journal is © The Royal Society of Chemistry 2024